aie26

Topic 06 : Data Modelling

Part 01 : Data Modelling - Introduction

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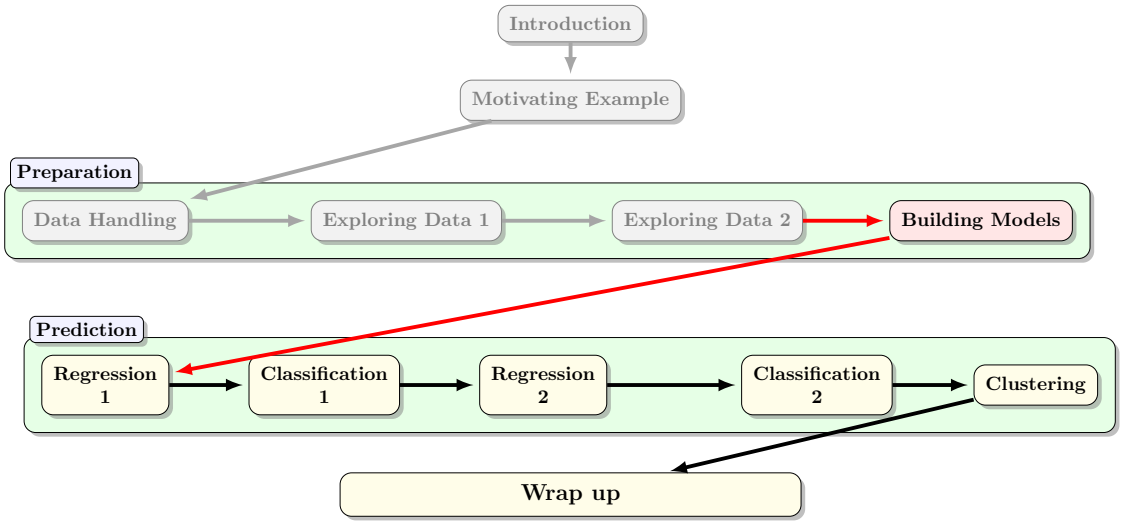
Autumn Semester, 2026

Outline

- Components of a machine learning problem
- Machine learning concepts and notation
- Bias vs variance

Wrap up

AI Engineering (Week 6)



Outline

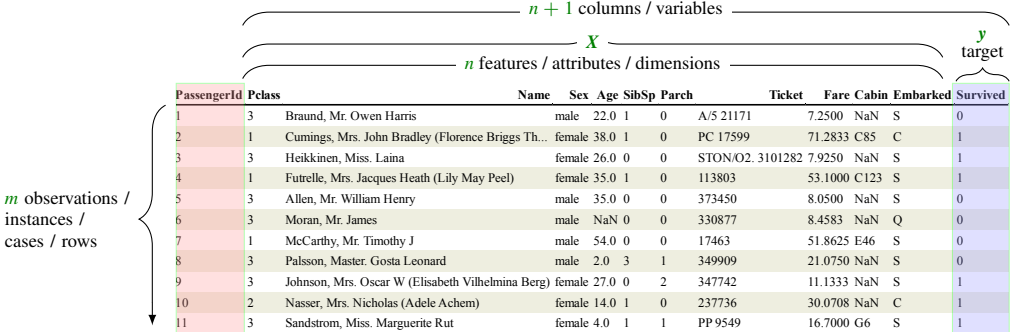
1. Machine Learning (ML) Overview 3
1.1. Review of terminology and notation 4
1.2. Components of a Machine Learning Problem 5
1.3. Problem–Task–Experience Perspective 10
1.4. Taxonomy of Machine Learning Methods 11
1.5. Statistical Models vs Machine Learning Models 13

2. Modelling Process 14
2.1. Models and error 16

Terminology / Notation

PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
1	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	0
2	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	1
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- A labeled dataset consists of m rows $\times$ $(n + 1)$ columns / variables.
- Use bold to represent vectors and matrices.

Terminology / Notation

$n + 1$ columns / variables

$\boldsymbol{X}$

n features / attributes / dimensions

$\boldsymbol{y}$ target

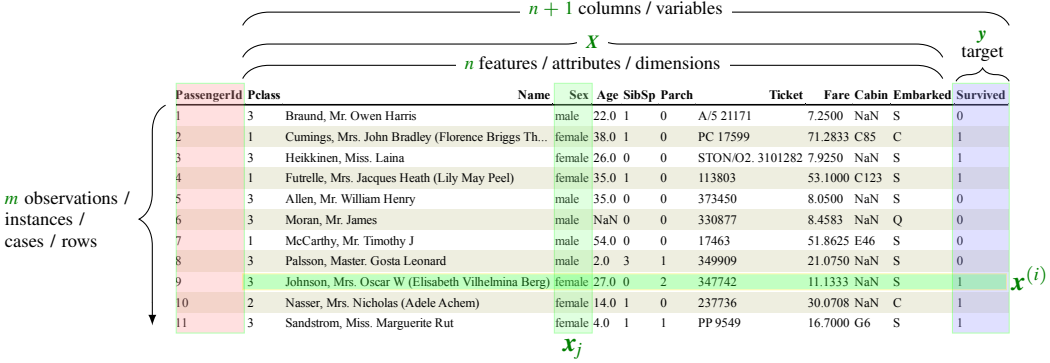
m observations / instances / cases / rows

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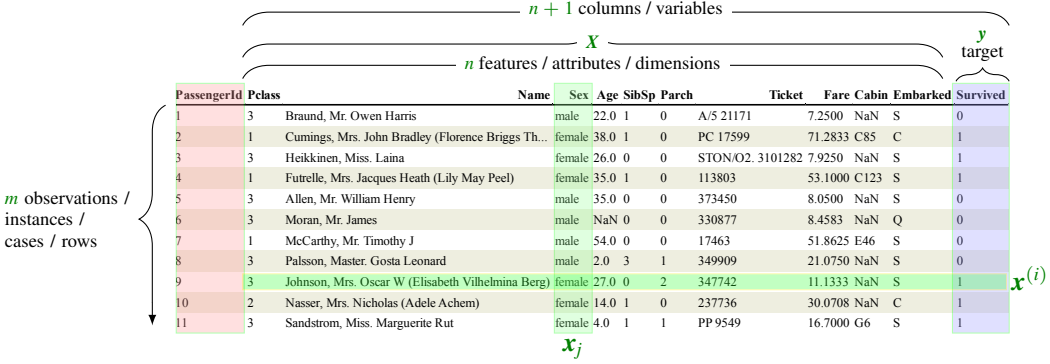
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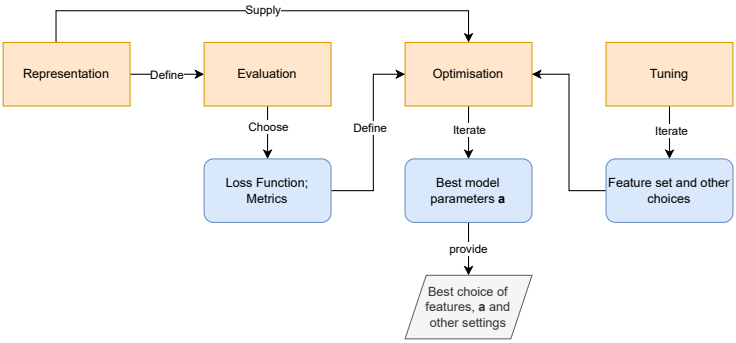
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- Use subscripts to indicate particular feature / attribute / column ..... $\boldsymbol{x}_j$
- Use superscript in parenthesis to indicate particular observation / instance/ case / row ..... $\boldsymbol{x}^{(i)}$
- So $x_j^{(i)}$ (or $x_{i,j}$) is the i-th observation in the j-th feature ..... $x_j^{(i)}$

Components of a Machine Learning Problem

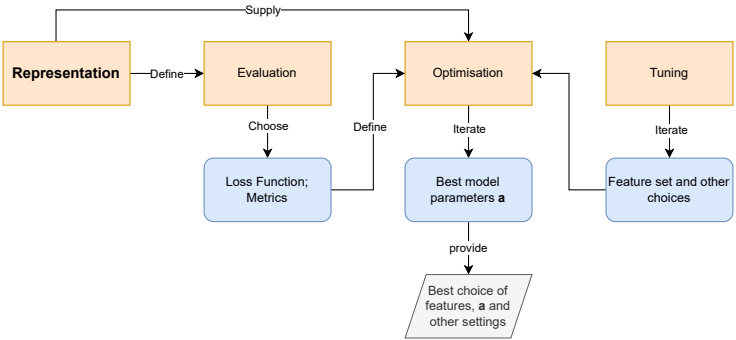
Where do we start????

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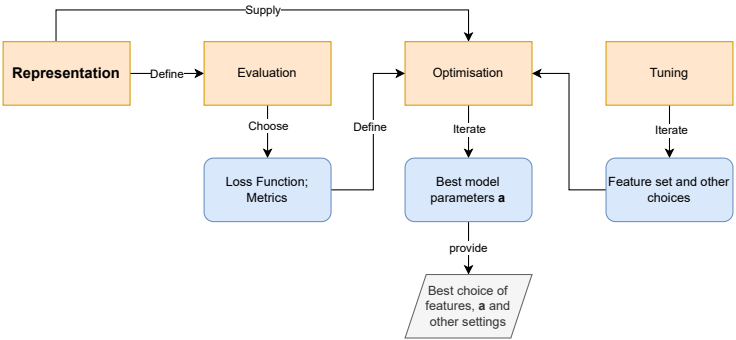
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Component 1: Representation

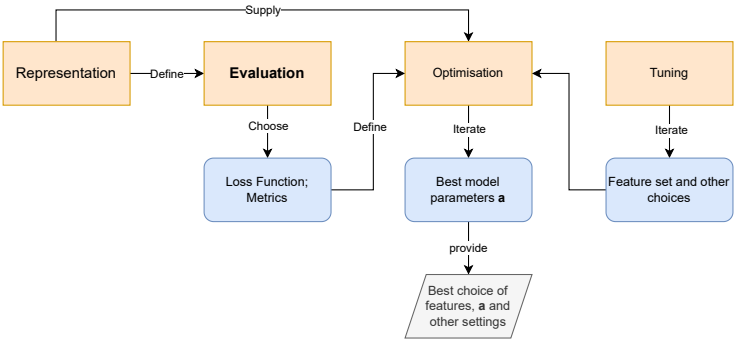


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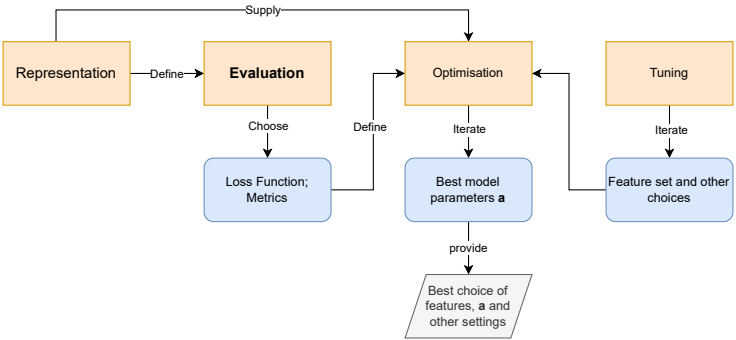


- Given the business objective and the data, choose what type of ML problem to solve: regression, classification, clustering, etc.
- Choose an algorithm for that problem. For example, if the ML problem is classification, we can choose KNN (other techniques will be seen later)
- What is the target and what features are available?
- These choices are made once - they are foundational for what follows.

Component 2: Evaluation

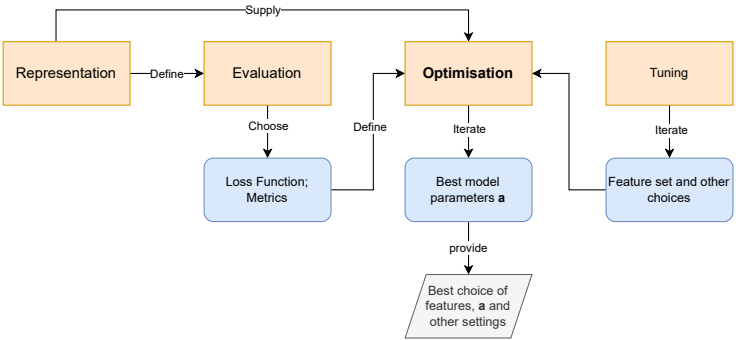


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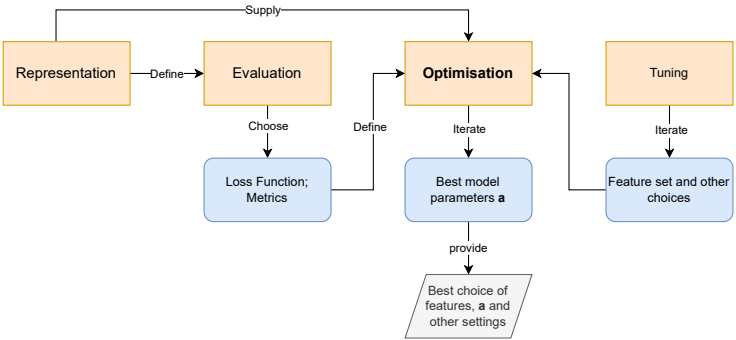


- Need an Objective Function that we can use to measure how well the model represents the data.
- For classification: how often does the model predict the right class, so that the predicted target and the actual target match each other?
- For regression: how close are the predicted target values to the actual target values?
- Generally the objective function takes actual and predicted target values and returns a single nonnegative number.

Component 3: Optimisation

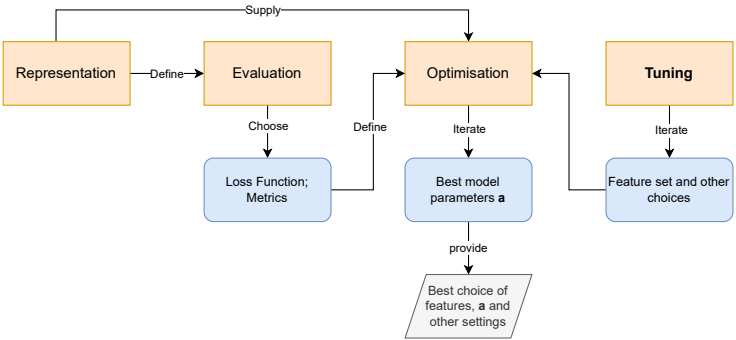


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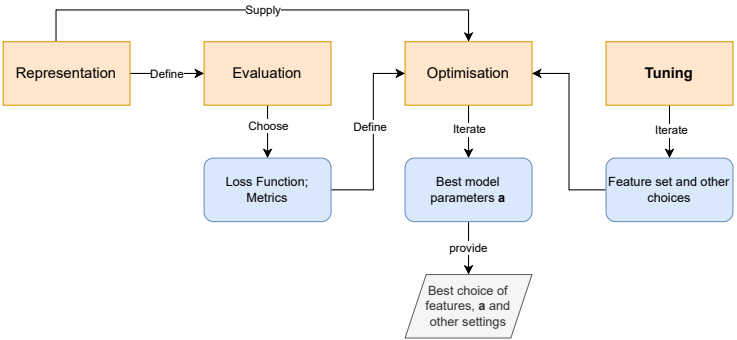


- Most ML algorithms are *iterative*: start with a guess at the model parameters $\boldsymbol{a}$, and improve them until the process terminates.
- The nature of $\boldsymbol{a}$ and how the code searches for better $\boldsymbol{a}$ depends on the Representation, refined by the Tuning choices - we will see examples later
- The choice of when to terminate the optimisation procedure depends on the Evaluation choice.
- Most of the computational resources are needed for the Optimisation component.

Component 4: Tuning



Component 4: Tuning



- Tuning is also iterative and is applied on top of Optimisation
- Tuning parameters (also known as *hyperparameters*) are kept constant during Optimisation
- Tuning parameters depend on Representation. For example, we have seen one hyperparameter already: k in the KNN classifier.
- For Tuning, GridSearch (exhaustive over a range) and RandomSearch (pick represntative values) are popular ways to search for the best hyperparameter values.

Data Modelling (aka Machine Learning)

As alternative to the four component (Representation / Evaluation / Optimisationi/ Tuning) viewpoint we can think of a machine learning problem as

Definition 1 (Machine Learning)

Study of algorithms that improve their performance P at some task T with experience E.

Well defined learning task: $< P, T, E >$

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- Task type: classification, regression, ...
- Linear vs nonlinear?
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- How many historical observations are needed?
- How accurate/noisy is the data?
- Do we have missing values?
- Is the data representative?

Taxonomy of Machine Learning Models . . .

. . . by Intuition/Motivation

. . . by Algorithmic Properties

. . . by Fixed/Variable Number of Parameters

Taxonomy of Machine Learning Models . . .

. . . by Intuition/Motivation

- Geometric models use intuitions from geometry such as separating (hyper-)planes, linear transformations and distance metrics.
- Probabilistic models view learning as a process of reducing uncertainty, modelled by means of probability distributions.
- Logical models are defined in terms of easily interpretable logical expressions.

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. . . by Algorithmic Properties

- Regression models predict a numeric output.
- Classification models predict a discrete class value.
- Neural networks learn based on a biological analogy
- Local models predict in the local region of a query instance.
- Tree-based models (recursively) partition the data to make predictions.
- Ensembles learn multiple models and combine their predictions.

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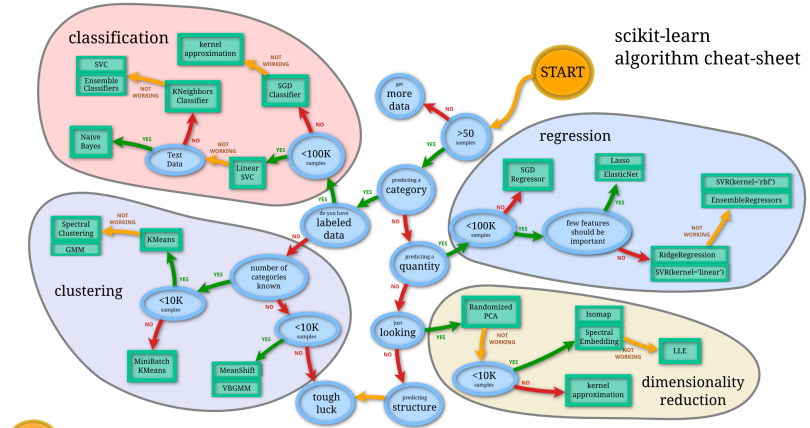
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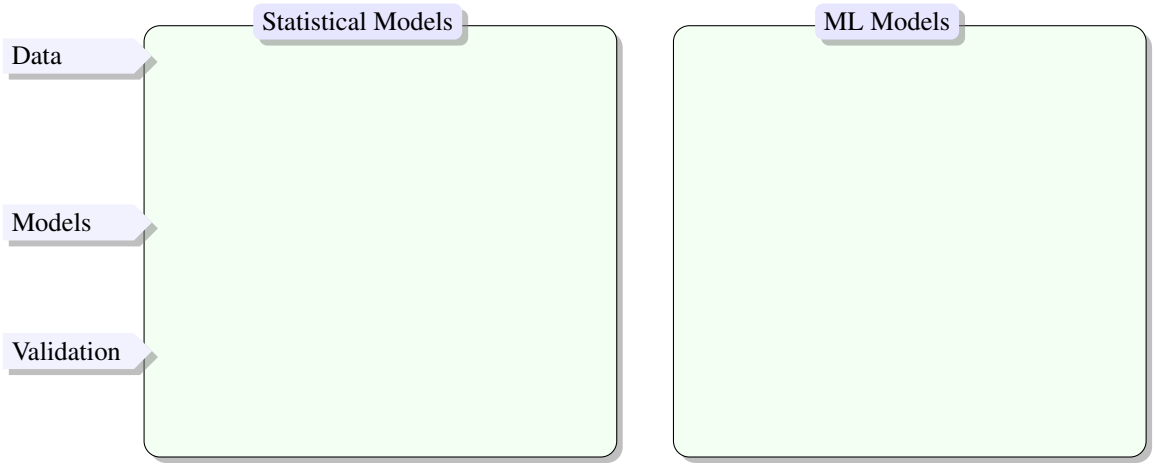
- Parametric models have a fixed number of parameters.
- In non-parametric models the number of parameters grows with the amount of training data.

Aside: Scikit-learn Flowchart of Models (Shallow Learners)



A neural network with more than one hidden layer is called a deep learner, all other learners are shallow learners.

Statistical Models vs Machine Learning Models



Statistical Models vs Machine Learning Models

	Statistical Models	ML Models
Data	• Usually small (< 1000 observations) • Low dimension (< 10 variables) • Can have detailed understanding of data • Data is clean — human has looked at each data point	• Can be huge (million+ observations) • Large dimension (1000+, more for vision) • Too large for human to parse / understand • Data not clean — humans can't afford to understand/fix each point
Models		
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Splitting data into train+test(+validation) is vital

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The Pipeline Metaphor

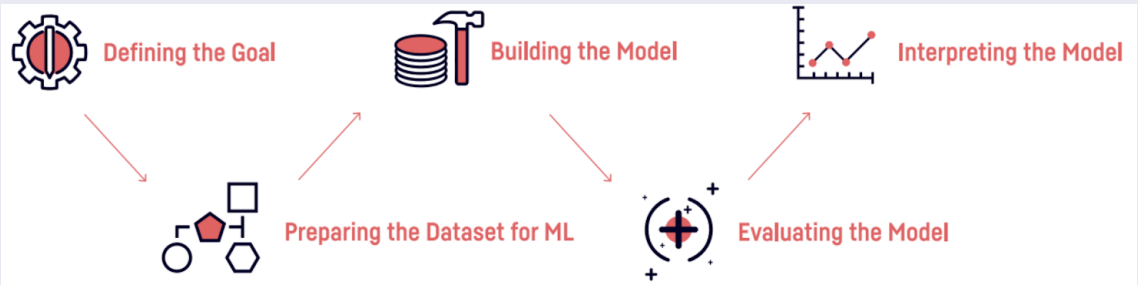
Model Building Pipeline



Source: Dataiku

The Pipeline Metaphor

Model Building Pipeline



Source: Dataiku

Comments

- We saw the first two stages in previous weeks
- This week we look at the remaining stages
- Of course this pipeline is a simplification. In reality it is iterative.

What does a (supervised learning) model look like?

Definition 2 (Linear Model)

General form of linear model used in this module looks like

$$y_i \sim f_i^{(1)} + f_i^{(2)} + \ldots + f_i^{(\mathrm{n})}$$

where y_i is the value of the response variable (target) for row (observation) i, and $f_i^{(\mathrm{j})}; j = 1, \ldots, n$ is the value of the jth feature for that observation.

In words: each target value y_i in the data is modelled as a linear combination of the model parameters $\boldsymbol{a}$ and the features, plus some error.

What does a (supervised learning) model look like?

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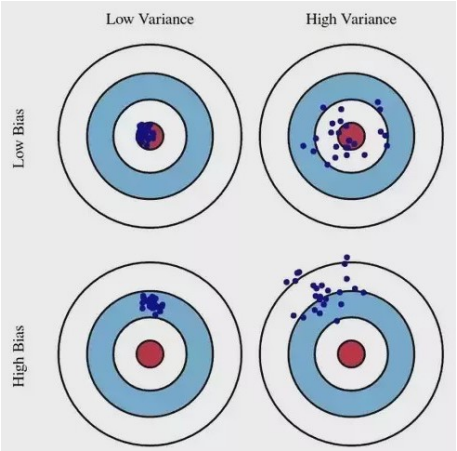
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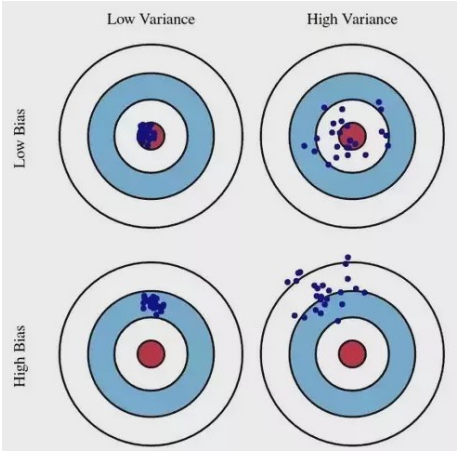
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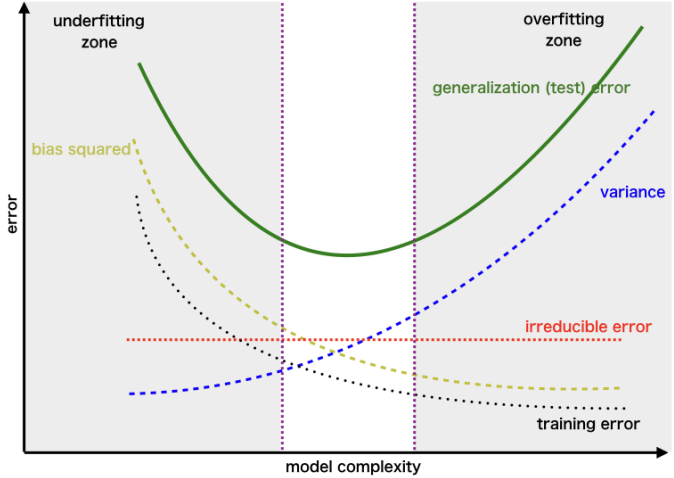
The goal of modelling is to find a so that the *prediction error* (loss function $\sim \|\varepsilon\|$) is a minimum.

Bias-Variance and Total Error

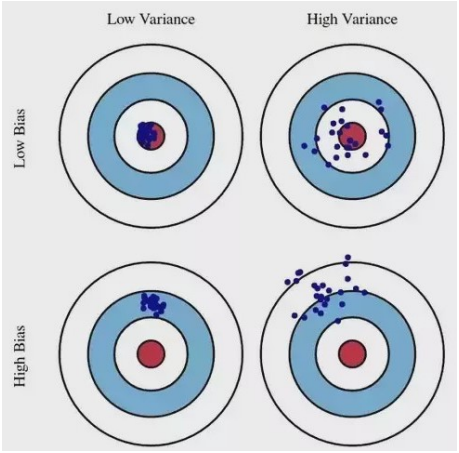


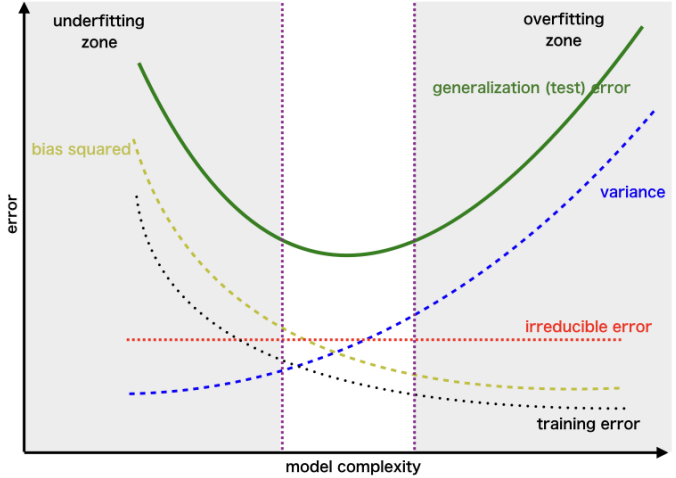
Bias-Variance and Total Error





Bias-Variance and Total Error





Look for parameters ***a*** that minimise the generalization error (estimated using the test set that was not used during training)

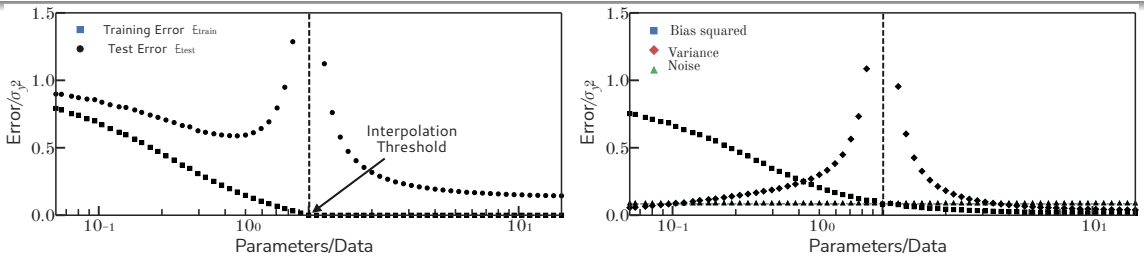
Introducing Double Descent

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Introducing Double Descent

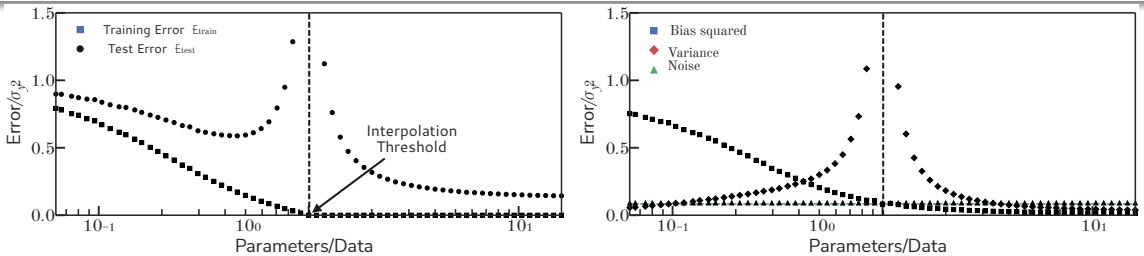
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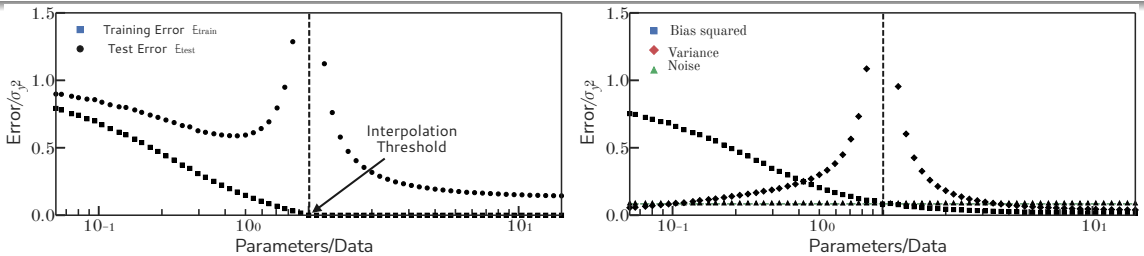
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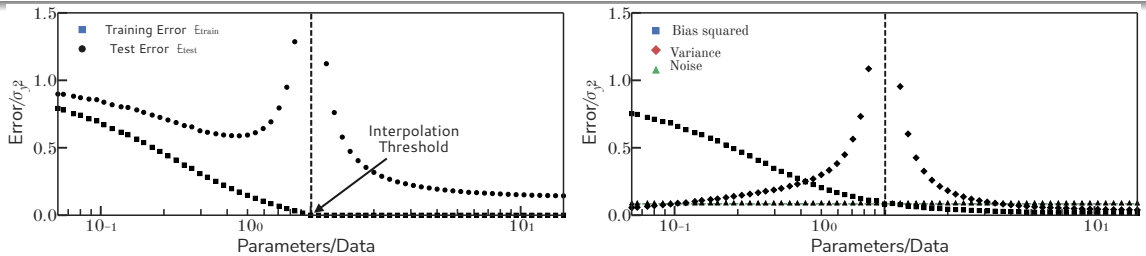
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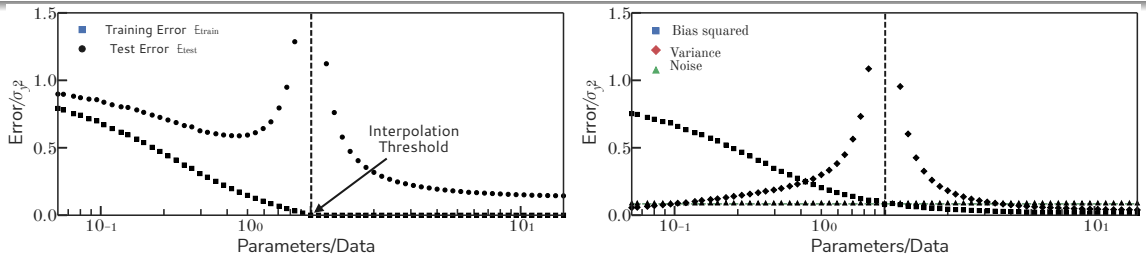
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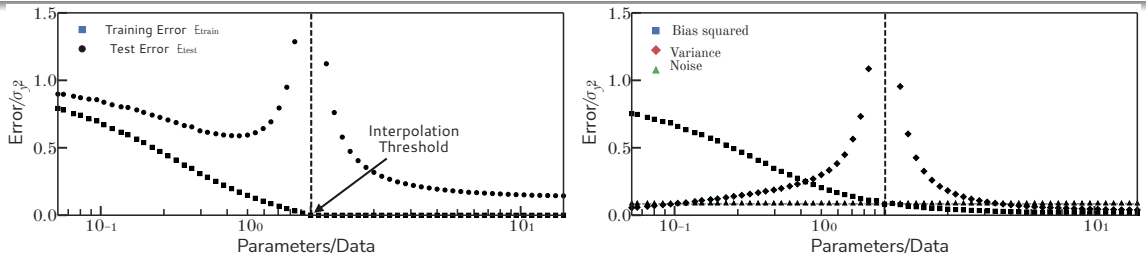
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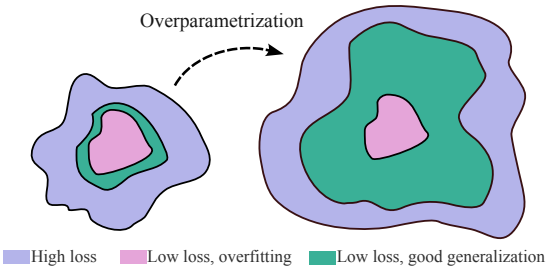
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- For deep learning - see AI Engineering 2 (semester 2)

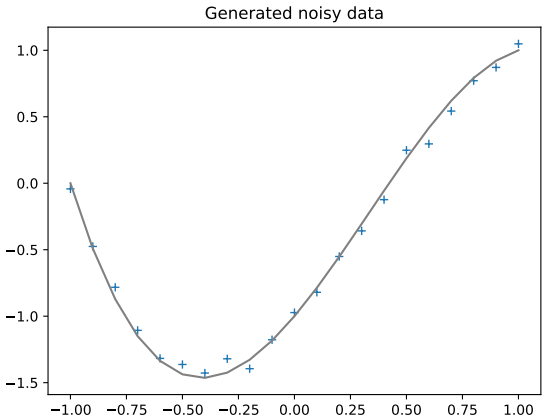
Effects of Double Descent



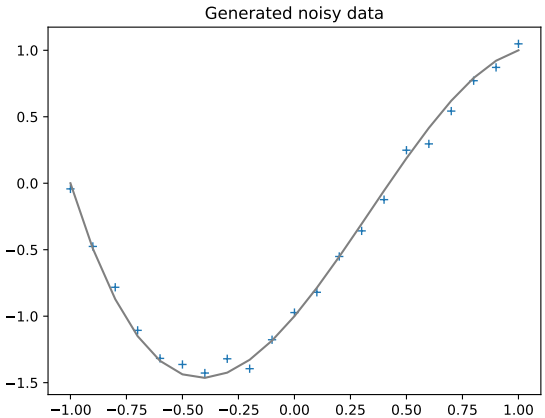
Extending to high dimensions

- Conventional models are usually not over parametrised: the number of features (columns) is usually much less than the number of cases (rows), i.e., the feature matrix is long and narrow
- Since double descent does not occur, the middle optimal settings zone (indicated by green colour adjoining) is narrow
- With deep learning, overparameterised models are easier to set up, so double descent allows a much wider optimal settings zone.
- This also allows for smaller bias without dramatically larger variance.

Bias-Variance Example: Noisy data



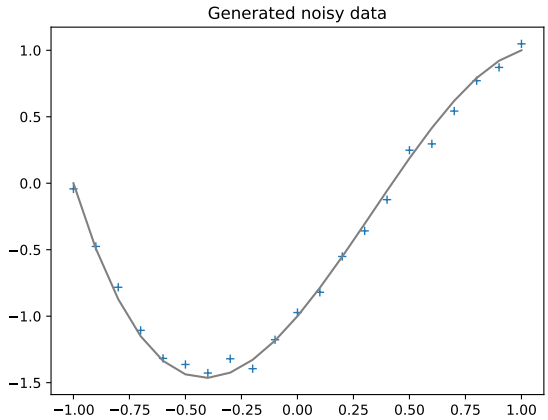
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Comments

- Given data with some error (noise)
- Expected underlying model is indicated by the grey curve
- In the next slides we will compare different models, indicated by red curves
- The models have different numbers of *features*
- The values predicted by each model lie on the red curve
- The loss function is an estimate of how much the grey and red curves differ

Bias-Variance Example: Noisy data

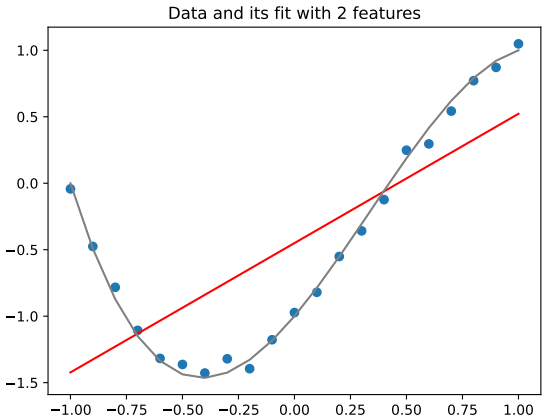


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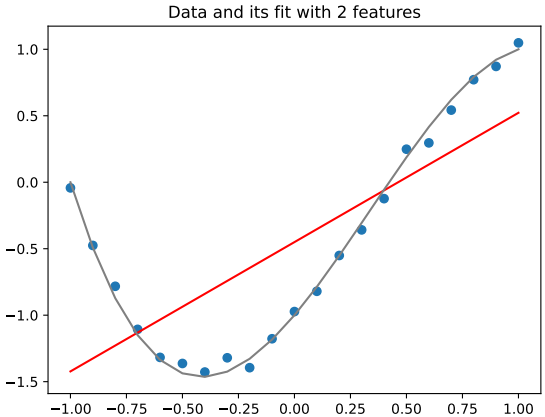
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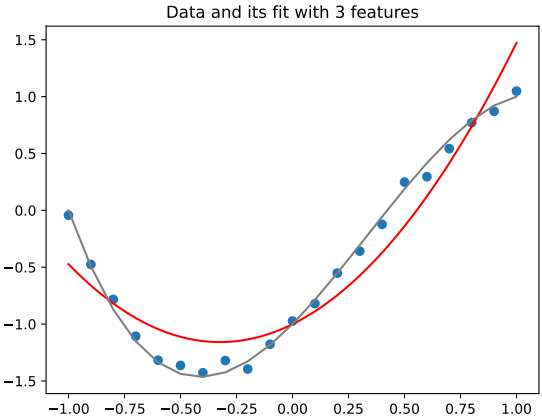
Look for the number of features that minimise the loss function

High Bias, Low variance

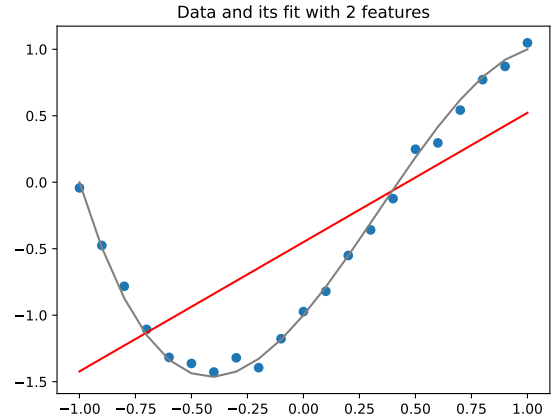


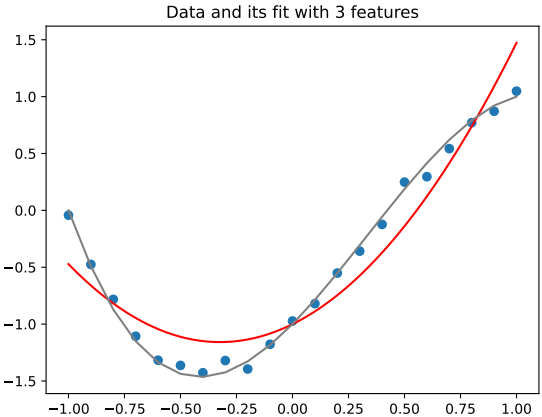
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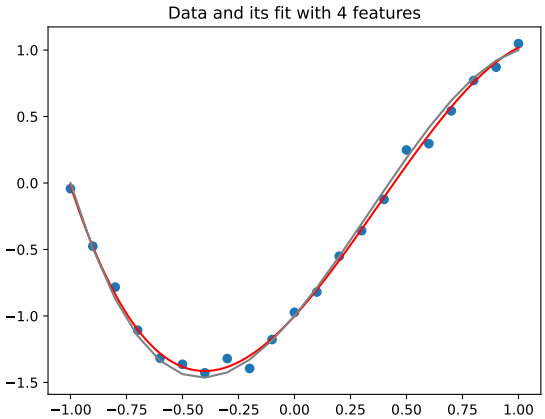
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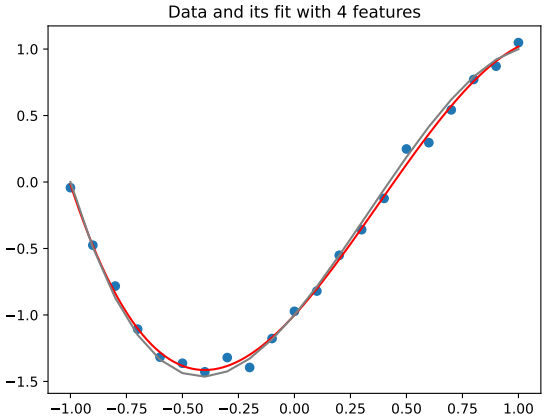


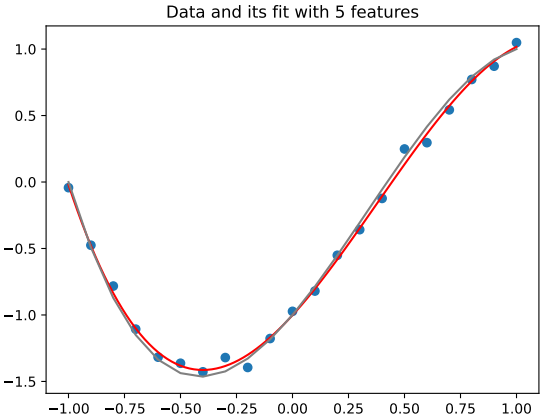
Need more features: add more - but which ones?...

Low Bias, Low variance

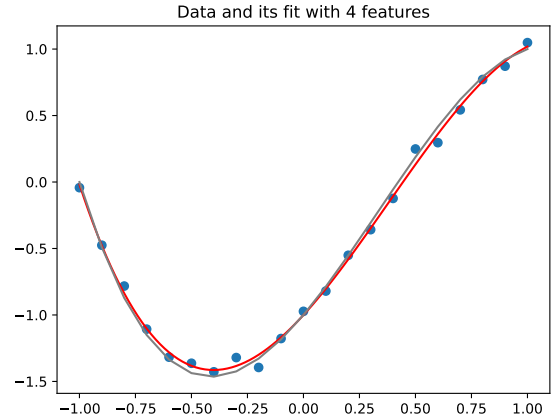


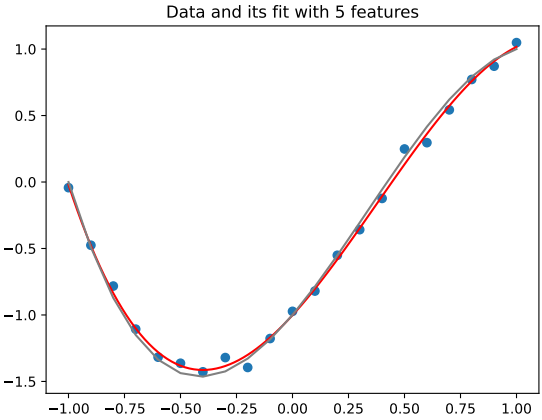
Low Bias, Low variance





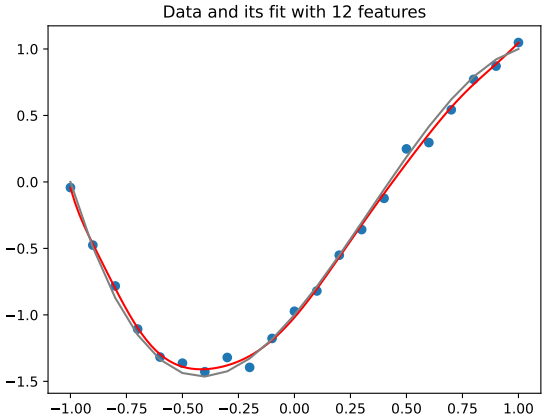
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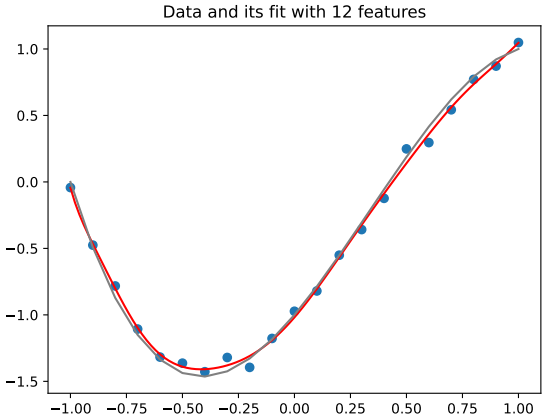


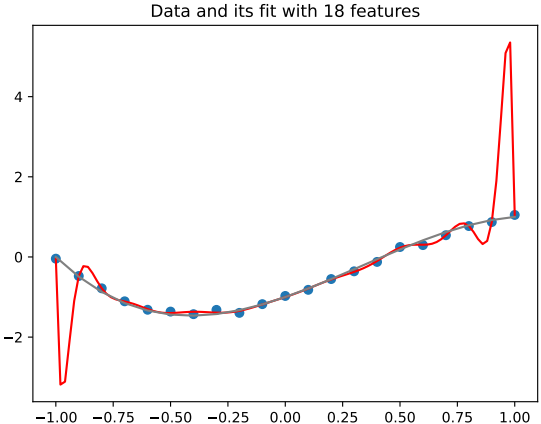
About the right number of features…

Low Bias, High variance

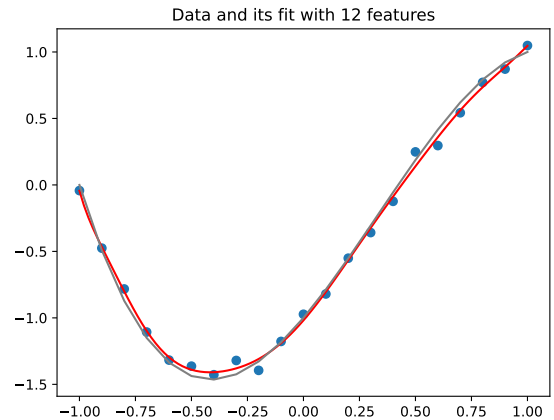


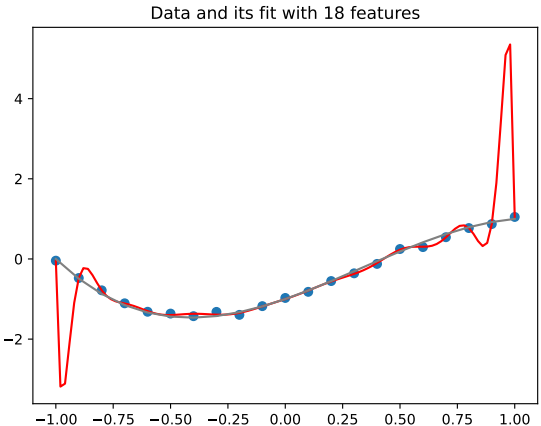
Low Bias, High variance





Low Bias, High variance





Too many features: remove some, but which ones?...

Example Model Types

Model	Applications	Concerns
Logistic Regression	X-ray classification	Regression with transformed variable
Fully connected networks	Classification	Classical ANN: choose encoding and size
Convolutional Neural Networks	Image processing	deep learning - choose segmentation
Recurrent Neural Networks	Voice recognition	ANN with feedback - how much?
Random Forest	Fraud Detection	Ensemble method - how many?
Reinforcement Learning	Learning by trial and error	Choose goal and penalties
Generative Models	Text, Image creation	Choose parameters
K-means	Segmentation	Choose distance function and k
k-Nearest Neighbors	Recommendation systems	Choose distance function and k
Bayesian Classifiers	Spam and noise filtering	Deal with imbalances

Before you start...

Does a *pre-trained* model exist?

Before you start...

Does a *pre-trained* model exist?

Transfer Learning

- Building a model from scratch is resource-intensive
- Open source data and model exist, particularly for deep learning (not in this nmodule)
- Most frameworks provide example models that can be used as a template
  - Select a similar model
  - Prune it (remove unnecessary terms)
  - Train using the pruned model as a starting point

When model training is very expensive, such as training a LLM, foundation models provide a good starting point. Hugging Face provides a wide selection of such models.